25901874 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

second SEMESTER

Skill Development Components (SDC) - MG2SDCSNP100 - ORIENTATION TO PHYSIOTHERAPY AND EXERCISE PHYSIOLOGY

(2025 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Mention the importance of methods of assessing body composition. [K] / [CO1]
2. Mention one example related to Exercise Physiology. [K] / [CO1]
3. State the importance of factors influencing body composition. [K] / [CO1]
4. Name two fuels used in the aerobic energy system. [K] / [CO3]
5. Write one practical use of Assessment of cardiorespiratory fitness in sports. [A] / [CO3]
6. What happens to cardiac output during an acute exercise? [K] / [CO3]
7. Give two examples of Chronic adaptations in respiratory system. [K] / [CO3]
8. Mention two factors influencing muscular endurance. [K] / [CO4]
9. State two benefits of resistance training. [K] / [CO4]
10. State two ethical responsibilities of a sports physiotherapist. [K] / [CO5]

11. State two components of injury prevention programmes. [K] / [CO5]

12. State two physiological effects of massage. [K] / [CO5]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

13. Discuss any two factors affecting energy generating capacity. [U] / [CO2]

14. Explain practical applications of Energy cost of sports. [A] / [CO2]

15. Explain the FITT principle in designing an aerobic training programme. [U] / [CO3]

16. Write a short note on RPE. [K] / [CO3]

17. Explain how exercise enhances neuromuscular coordination. [U] / [CO3]

18. Explain the role of rest and recovery in training adaptation. [U] / [CO4]

19. Explain the different phases of a training programme. [U] / [CO4]

20. Explain the importance of physiotherapy in injury prevention. [U] / [CO5]

21. Describe the types and therapeutic uses of TENS. [U] / [CO5]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

22. Explain the Phosphagen, Anaerobic Glycolytic, and Oxidative systems and their importance in sports performance. [U] / [CO2]

23. Differentiate between aerobic and anaerobic training based on their physiological basis and training characteristics. [An] / [CO3]

24. Discuss the importance of skill-related fitness in athletic performance and daily functional activities. [U] / [CO4]

25. Explain the principles, techniques, and therapeutic benefits of hydrotherapy. [U] / [CO5]

[10x2 = 20]